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# Resistor Combination Calculator
# Calculates total resistance in series and in parallel.
# Own twist: also uses Ohm's law to find the current for a supply voltage.


def get_positive_number(prompt, whole_number=False):
    """Keep asking until the user types a valid number above zero.
    If whole_number is True, the input is converted with int(),
    otherwise it is converted with float()."""
    while True:
        try:
            if whole_number:
                value = int(input(prompt))
            else:
                value = float(input(prompt))
            if value > 0:
                return value
            print("Error: the number must be greater than zero. Try again.")
        except ValueError:
            print("Error: please type a valid number, not letters. Try again.")


def series_resistance(resistors):
    """Return the total resistance when resistors are in series."""
    total = 0
    for r in resistors:  # add every resistor value together
        total = total + r
    return total


def parallel_resistance(resistors):
    """Return the total resistance when resistors are in parallel."""
    reciprocal_sum = 0
    for r in resistors:  # add 1/R for every resistor
        reciprocal_sum = reciprocal_sum + (1 / r)
    return 1 / reciprocal_sum  # total = 1 / (1/R1 + 1/R2 + ...)


def show_current(label, voltage, resistance):
    """Own twist: use Ohm's law (I = V / R) and print the current."""
    current = voltage / resistance
    if current < 1:
        # small currents are easier to read in milliamps
        print(f"  Current in {label} circuit: {current * 1000:.2f} mA")
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    else:
        print(f"  Current in {label} circuit: {current:.2f} A")


def main():
    print("=== Resistor Combination Calculator ===")

    while True:  # loop lets the user do many calculations
        # Ask how many resistors, then collect each value in a list
        count = get_positive_number("\nHow many resistors do you have? ", True)
        resistors = []  # list that stores all resistor values
        for i in range(count):
            value = get_positive_number(f"Enter value of resistor {i + 1} (ohms):
            resistors.append(value)

        # Calculate both totals using the functions
        total_series = series_resistance(resistors)
        total_parallel = parallel_resistance(resistors)

        print("\n--- Results ---")
        print(f"Resistors entered: {resistors}")
        print(f"Total resistance in SERIES:   {total_series:.2f} ohms")
        print(f"Total resistance in PARALLEL: {total_parallel:.2f} ohms")

        # Decision: give a helpful note depending on how many resistors
        if count == 1:
            print("Note: with one resistor, series and parallel are the same.")
        elif count == 2:
            print("Note: two resistors, the simplest combination.")
        else:
            print(f"Note: you combined {count} resistors.")

        # Own twist: Ohm's law for a supply voltage
        voltage = get_positive_number("\nEnter supply voltage in volts: ")
        show_current("series", voltage, total_series)
        show_current("parallel", voltage, total_parallel)

        # Ask if the user wants to go again
        again = input("\nCalculate again? (y/n): ").strip().lower()
        if again == "y":
            continue
        elif again == "n":
            print("Goodbye!")
            break
        else:
            print("Not a valid choice, so the program will close. Goodbye!")
            break
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main()
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